

WHAT DO DOULAS DO?

EVIDENCE FROM MEDICAID DOULA COVERAGE

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Abstract

Doulas, or specifically birth/labor and delivery doulas, have long been viewed as a useful support in the labor and delivery process — particularly in cases in which a delivering family may not know when to advocate for themselves or feel comfortable doing so. Yet little is known about population-level effects of doulas in deliveries for low-income families. This paper studies the impact of Medicaid coverage of pregnancy- and labor-and-delivery-related doulas. I find that after the implementation of state policies to provide Medicaid support for doula services, the number of registered doulas in a state increases and the likelihood of delivery by Cesarean section drops. Overall, the rate of C-sections drops by about 3.8%, but drops by as much as 5 and 7% for Black and Hispanic mothers respectively. Effects are driven by a reduction in failed labor attempts.

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1 Introduction

Maternal and infant mortality are high, and seem to be increasing, in the United States (Hill et al., 2023). When comparing birth-related outcomes, the US does quite poorly relative to other countries — and does significantly worse for people of color than for white people giving birth. For example, if the U.S. Maternal Mortality Rate (MMR) for Black mothers were separated from the rest of the population and compared to other countries, it would be in the top-100 countries in a ranking of MMRs — meaning that Black mothers in the US experience mortality rates (55.3 deaths per 100,000 live births) similar to women in Panama (50) and Mexico (59), lagging far behind even some of our geographic neighbors, like Cuba (39) or Puerto Rico (34); and astonishingly behind our economic peers, like the U.K. (10), France (8), or Germany (4) — despite the fact that Americans pay more for health care per capita than those in any other wealthy country (CIA, 2023; Gunja et al., 2023). Health professionals and policymakers alike have grappled with the question of how to reduce maternal and infant mortality in the US — particularly for those most vulnerable to maternal mortality, women of color. One potential strategy to improve these outcomes is to provide doula care to all pregnant women. Doulas are professionals who support people through major health transitions — for birth, doulas provide emotional, physical, and informational support and assist mothers with communication and advocacy with the broader birth health team. Doulas have helped women identify health warning signs, such as high blood pressure, that could be deadly if left unnoticed. Women who gave birth with doulas present were also less likely to deliver low-birthweight babies, experience birth complications, require a cesarean section delivery, or experience postpartum depression (Gruber et al., 2013; Falconi et al., 2022). Doulas can help inform mothers-to-be and can bridge linguistic and cultural gaps between mothers and their doctors.

Unfortunately, many of the women most at risk for birth-related complications lack information or resources to connect with doulas. This leaves them potentially at risk, unaware of their precarious state, and unable to effectively advocate for their health. To improve

birth-related outcomes, some states have begun to reimburse doula fees for Medicaid patients — such as Oregon in 2013, Minnesota in 2014, Florida in 2019, California in 2021, and Michigan in 2020, among others.¹ In this project, I aim to analyze national birth data to determine whether access to doulas has improved health outcomes for women and infants.

To answer this question, I will compare outcomes for births that were paid for with Medicaid across states that did and did not reimburse doula services. Additionally, I'll compare Medicaid-covered births to non-Medicaid covered births in reimbursing and non-reimbursing states over time, to see whether the doula reimbursement improved any existing income-gap in birth outcomes. Finally, I will assess whether doula reimbursement reduced infant and maternal morbidity gaps across racial and ethnic groups. For each of these analyses, I will use county of residence information and mother demographics, as well as birth-related health outcomes and method of payment, from the National Center of Health Statistics' Vital Statistics data. Early results, comparing Medicaid births across states over time, are somewhat less likely to experience severe maternal morbidity, very low birthweight and very low APGAR scores, and these births are less likely to result in hospital transfers or infant admittance to the NICU. To truly get at the heterogeneity of any effects – and to see how the broad policy change is actually impacting different groups of people — I will conduct subgroup analyses based on race/ethnicity, individual demographics, and local demographics. Due to the variety in start dates for these states' Medicaid coverage, I will follow current approaches to ensure the dynamic nature of this study is taken into consideration, likely using the Callaway-Sant'Anna approach of inverse probability weighting or doubly-robust estimands and Andrew Goodman-Bacon's estimator decomposition (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021). The rest of this paper is organized as follows: Section 2 briefly discusses economic work on birth-related health outcomes, defines doulas, and describes some detail about Medicaid coverage of doula care. Section 3 describes the

¹At the time of this draft, 8 states and the District of Columbia have formally adopted Medicaid coverage of or reimbursement of birth doula care. Only Oregon, Minnesota, Florida, and New Jersey began to actually cover doula care prior to the end of observable data (Chen, 2022).

data and empirical strategy used in these analyses. I discuss the results of the main specification, as well as potential predictor variables. Finally, I summarize my findings so far in Section 5 and provide a brief discussion of policy implications and next steps in Section 6.

2 Background

2.1 Birth-Related Health Outcomes & Disparities

While the U.S. is a leader in medical spending and innovation, it has failed to maintain improvement in birth-related health outcomes achieved by the 1990s and continues to lag behind European peers (Chen et al., 2014). In several states, maternal mortality has actually increased since the 1990s — in some cases, the number of maternal deaths per live birth have actually surpassed the same statistic from the 1960s in recent years.² While these statistics are poor across all dimensions (even the most privileged in the United States still face high risks in labor and delivery relative to similarly-privileged peers in other countries), they are especially strong for groups facing some sort of barrier to health care — whether that barrier comes through income, health insurance status, education, racial discrimination, language barriers, or even geographic location. Economists are not oblivious to these issues, and much work has been done to understand possible mechanisms behind these terrible outcomes. First, a branch of literature discuss the history of labor and delivery in the United States, documenting the transition from home births to hospital births and genuinely impressive improvements in infant and maternal mortality over time (Troesken, 2008; Moehling and Thomasson, 2014; Anderson et al., 2022; Hollingsworth et al., 2022; Jacks et al., 2021; Jayachandran et al., 2009; Cutler et al., 2012; Collins and Thomasson, 2004; Haines, 2011; Anderson et al., 2020). Sanitation and medical innovation are credited as contributors to the historic improvements in the early-to-mid 1900s (Moehling and Thomasson, 2014; Clay

²Using maternal death data from Iowa and Pennsylvania, there is a clear increase in maternal deaths per total live births in both states — the statistic for 2021 in Iowa was more than double Iowa’s 1960 stat (20.38 maternal deaths per 10,000 live births in 1960 compared to 48.93 per 10,000 live births in 2021).

et al., 2023, 2024) and many studies have examined different potential solutions to poor health outcomes in more modern times (Altındağ et al., 2024; Geruso and Spears, 2018b,a).

Since these health disparities are most pronounced along racial and ethnic lines — and given the U.S.’s history of systemic racism in medicine, particularly toward Black Americans — bias in healthcare is considered by many to be a likely contributor to the problem. Work has been done to document the existence and consequences of bias in medicine and to assess approaches to reduce bias in health systems, finding that reducing provider bias can improve patient experiences and outcomes (Eli et al., 2019; Wagner et al., 2023; Dominguez et al., 2008; Carty et al., 2011). Even in cases where bias does not seem to be driven specifically by racist beliefs (or along racial lines), evidence consistently shows that medical actors behave differently for different patients for reasons *outside* of actual medical differences, suggesting medical bias may have a firm foothold in the current system (Currie et al., 1995; Johnson and Rehavi, 2016; Currie et al., 2024; Eli et al., 2019; Yearby, 2021).

Finally, patients are aware — to some degree — of the way their personal presentational intersections may negatively impact their health outcomes. Black and Indigenous birthing people have commented on their likelihood of and experiences with race-based medical mistreatment (Nuru-Jeter et al., 2009; Nguyen et al., 2023; Thackrah et al., 2021). Patients aware of racial or other biases in medicine may choose not to trust the advice of their providers — as can be seen even in long-term, intergenerational studies evaluating the effects of the Tuskegee Study’s abhorrent dishonesty (Shavers et al., 2000; Gamble, 2016; Alsan and Wanamaker, 2018). Birth workers have expressed outrage and demanded change within their profession to improve these outcomes and reduce the role of racism and bias in their patients’ experience — often calling for improvement in culturally-relevant and trauma-informed care (Thackrah et al., 2021; Louise-Jacques, 2024).

2.2 Doulas

In order to understand what sort of effects we might reasonably expect from doula coverage, it is crucial to understand what a doula is — and, just as importantly — what a doula is not. Doulas are trained and (typically) certified care providers who assist and support people through a wide variety of health transitions, from fertility through delivery and postpartum all the way through later-in-life health crises and end-of-life. Doulas are increasingly supporting a diverse range of clients: some specialize in assisting Black or indigenous people, some specialize in gender diversity and gender transition processes, and some focus on high-risk pregnancies and the provision of trauma-informed care. In any setting, doulas provide physical, emotional, and informational support to their clients. Clients of doulas benefit from the wealth of knowledge and experience the doula has with the relevant transition — which, in many cases, the client is experiencing for the first time. While the breadth of available doula services is wide, the most common type of doula care is connected to pregnancy and delivery. In fact, unless clarified to be a different type of doula, the word ‘doula’ consistently evokes an image of a labor and delivery supporter. Through the rest of this paper, ‘doula’ will be used to describe pregnancy-related doulas — most often, the doulas present in the labor and delivery room.³

Doulas are trained, experienced, and certified providers, but there is currently no uniform training process or certifying/licensing board in the United States. This lack of a uniform process makes it difficult to understand exactly how many doulas are practicing in the U.S. at any given time, though estimates suggest approximately 9,000 doulas practiced in recent years (Reed and Goldman, 2023). At the same time, there have been roughly 3.6 million

³As an important aside, the term ‘doula’ deserves some awareness. Doula, pronounced ‘doo-luh’, comes from an ancient Greek word *doulē*, meaning female servant (Center, 2023). This language choice is debated among actual doula practitioners; some find its continued use to be perpetuation of racist stereotypes (especially since many doulas are people of color) and forces a gendered perspective of this work, while others find it empowering to reclaim the word (Yiu, 2024; Salinas et al., 2022). At this time, a consensus has not been reached, and ‘doula’ remains the most commonly-used term for the type of labor and delivery worker being referenced in this project. As such, I will continue to use ‘doula’ throughout this paper, but will update terminology as our language evolves.

births per year — doulas only cover a very small share of births in this country. Doulas differ from other types of labor and delivery workers in many ways: training is shorter, certification is less formal, and licensing is inconsistent for doulas (Wallis and Bassham, 2023; Guenther et al., 2022). In fact, there is nothing to stop a person from claiming doula status in another person’s labor and delivery process — though reimbursement for doula services does typically require some form of certification.⁴ Doulas are not able to perform medical tasks, unlike certified nurse midwives (or certified nurses, PAs, or MDs). Doulas have, in general, more training and experience in labor and delivery room settings than their clients (and are not personally experiencing the difficulty of childbirth) who are able to contextualize the birthing person’s experience and assist the client in self-advocacy. Doulas may also help birthing people stick to their intended birth plan — especially in marginal cases for possible C-sections.

While evidence is somewhat limited, doulas seem to improve the birthing experience for their clients: when randomly assigned to have a doula, birthing people experienced shorter labor, increased likelihood of spontaneous vaginal deliveries, and decreased use of pain medication and Cesarean sections (Bohren and Cuthbert, 2017). Additionally, babies born with doula support were less likely to have low APGAR scores and birthing people were less likely to experience postpartum depression (Bohren and Cuthbert, 2017; Dekker et al., 2013).

2.3 Medicaid Coverage of Doulas

Despite doulas being a potential channel through which birth outcomes for the most disadvantaged could be improved, doula services are not always covered by private or public insurance, making them potentially inaccessible to those who need their services the most. In an effort to remedy this issue, several states and local groups have worked toward providing

⁴In Wallis and Bassham (2023)’s article, they state that a family member could be listed as a doula, even without any doula training. In practice, this type of care would not typically be eligible for financial support from private or public insurance.

funding for low-income birthing people — and in some cases, low-income birthing people of particular communities, such as a particular race or ethnicity. Medicaid policy is determined at the state level, so there is variety in access to doula care for Medicaid-covered births by state and time. The first states to pass legislation providing doula coverage to Medicaid patients did so beginning in 2011 (Oregon), though the first year of actual benefits began in 2014 (Minnesota and Oregon). Since then, over half of all states have some form of a doula effort in existence (?).

Medicaid coverage or reimbursement of doula care does, theoretically, increase access to doula services for low-income birthing people, but there are a few key problems that remain unsolved. First, coverage of doula services does not necessarily mean doulas will be available in a given location. Even if services are covered, access to doulas would remain an issue if there are no practicing (or properly certified) doulas in the area — or if the demand for doula services far exceeds the local supply of these workers. Second, coverage of doula services does not guarantee a good match between client and doula. For a client to receive the maximum benefit of working with a doula, they would need to be matched to someone knowledgeable, respectful and aware of their potential concerns, and someone with whom they feel comfortable. Everyone has different needs and preferences, so poor matches are likely to exist between doulas and clients. If these poor matches are common, we might expect the benefits of doulas to be quite small. Third, even when there are sufficient doulas in the local market that good matches can be made, the presence of a doula throughout the delivery process cannot fully erase the negative outcomes and experience folks face — especially if the systemic nature of racism in healthcare, for example, has already had a compromising effect on their health. In other words, doulas may be able to help, but may not be able to ‘move the needle’ tremendously on the extreme outcomes. Finally, even in cases with reasonable availability of well-matched doulas, and in cases in which a client could receive immense benefit from a doula’s care, the coverage and reimbursement policies themselves can be limited in both the dollar amount and the timing of benefits. For example, Oregon’s initial

reimbursement for doula services was only \$75, which was certainly not enough. Similarly, any care that requires an up-front payment from the client to eventually be reimbursed by Medicaid could present a barrier to access for these low-income patients.

3 Empirical Approach

This section describes the data and empirical approach I use to estimate the causal effects of Medicaid coverage of birth doula care.

3.1 Data

To analyze the effect of Medicaid reimbursement policies on doula supply, I use the downloadable NPES NPI file housed by the Centers for Medicare and Medicaid Services (CMS). The downloadable file logs all National Provider Identifiers from 2005 - present. Any provider wishing to be reimbursed for their services through insurance (including Medicaid) must register for a provider identifier. Providers may have multiple certifications added to their NPI over time and the available file does not register the dates of each certification. A taxonomy code for doulas began to exist in late 2009 – since then, over 20,000 providers have listed ‘doula’ as at least one of their provider taxonomy codes. I focus on enumeration dates for providers listing doula as their primary provider taxonomy code.⁵ Looking only at providers stating that doula services are their primary provision, I observe 18,246 providers across the country in 2026.

To analyze the effect of doula care on birth-related health outcomes, I use the National Center for Health Statistics’ Vital Statistics data from 2012 – 2022. These data contain vast information on demographic and birth-related health outcomes for both the person giving birth and the infant born. They include geographic indicators (county of residence), prior pregnancy outcomes (number of prior births for which a child is still alive or is no longer

⁵I acknowledge that anyone who lists ‘doula’ and provides doula services, but does not identify this as their primary service will not be measured in this specification.

living, prior terminations), maternal health risk factors (such as hypertension or diabetes), maternal behaviors (such as prenatal care visits and counts of cigarettes smoked during each trimester), birth-related health outcomes for the mother (including unplanned hysterectomy, transfer to the ICU, or blood transfusions), demographic data (mother’s age, education, race and ethnicity), and the source of payment for the birth (public insurance, private insurance, other). Infant health outcomes are also included, ranging from the birth weight to the presence of congenital complications or defects to an indication for whether the infant was transferred to the NICU and whether the infant is alive at time of the birth record being taken.

To define treatment states and their start-of-treatment time, I use information regarding Medicaid changes by (Chen, 2022), which I verified by reading state legislation files regarding Medicaid coverages. Several states did implement changes to their Medicaid programs directly, while others have had non-Medicaid interventions (like doula pilot programs) implemented. I use the month of benefits beginning as the first potential month of treatment. Since many people interact with a doula only in the labor and delivery room (and not in prenatal experiences), it is reasonable to test for effects starting in the month in which doula care began to be reimbursed. However, we may not expect all eligible birthing people to be immediately aware of their eligibility for doula services, or we may expect any impacts of doulas to build over time (as people are able to get to know a doula prior to labor) — so evaluating dynamic treatment effects will be critical. Medicaid-paid births in states that do not (or do not yet) cover doula services will be considered ‘untreated.’ I consider measured effects to be an intent-to-treat effect, which is likely to be an underestimate of true treatment effects of doula care in the labor and delivery room. Surely not all Medicaid-eligible patients in treated states are working with birth doulas. Because I do not have a measure of presence of doulas in the delivery itself, I do not aim to establish a causal relationship between doula care and outcomes based on those who select doula services compared to those who do not. Instead, I measure the effect of a change in Medicaid coverage which supports doula care on

doula registrations and birth outcomes.

3.2 Identification Strategy: Event Study

I use a dynamic difference-in-differences approach to estimate the causal effects of Medicaid coverage of doula care. This approach exploits cross-state variation over time and controls for aggregate time shocks, as well as fixed differences across states over time. In order for this approach to be valid, it must be true that changes in birth-related health outcomes for comparison states provide a good counterfactual for the changes that would have been observed for treated states, if Medicaid coverage had remained unchanged. My approach to estimating the effects of Medicaid coverage of doulas on birth-related health outcomes corresponds to the following equation:

$$y_{ismy} = \sum_{k=-12}^{12} \theta_k Doula_{s,t-k} + \alpha_s + \alpha_m + \alpha_y + \epsilon_{ismy} \quad (1)$$

where y_{ismy} is the outcome of interest for a birth event i in state s , month m , year y ; $Doula_{s,t-k}$ is an indicator the number of months prior to birth event for which doula care was reimbursed in state s ; α_s are state fixed effects; α_m are month fixed effects; and α_y are birth-year fixed effects.⁶ All reported standard-error estimates are clustered on the state to account for correlation within states over time.⁷

I favor a method that uses individual birth outcomes for a variety of reasons — most importantly to allow for later-observations. The population data at the county-level that I typically use for reliable demographic data, SEER’s population statistics, are not available through 2022 at the time of this draft. Thus, any calculation based on county-level birth rates would necessarily rely on strong population assumptions. For event-study figures, I

⁶I have also used month of conception, calculated using gestational age at birth, to assign the number of months covered by doula care — results are quite similar and are available on request.

⁷I also drop any state that had a documented doula program that was not conducted through that state’s Medicaid (or did not include all Medicaid patients). However, due to a variety of undocumented (or unpublicized) efforts to improve birth-related outcomes in smaller localities, it is possible some of my ‘comparison’ observations are, in fact, treated.

restrict the dataset to only include birth events for which Medicaid was the primary payer.⁸ The running variable in the event study figures is the number of months between the birth event and the state’s doula efforts began (formal start dates for doula reimbursement or coverage) — an observation in the “3” category represents the effect of the state’s policy on the given outcome, when the policy came into action 3 months prior to the birth event. Similarly, an observation in the “-12” category represents the effect of the state’s policy on the given outcome when the policy began 12 or more months *after* the birth event. All estimated effects are referenced to births occurring one month prior to policy implementation, relative to Medicaid births in comparison states. If the outcome for Medicaid birthing people remained on its trajectory (relative to the outcome for Medicaid birthing people in other states) prior to state policy implementation, coefficients for the months prior to zero should be statistically indistinguishable from zero — which would support (though not perfectly ‘prove’) the parallel trends assumption. Then, any coefficients that are statistically different from zero in the post period should represent the causal effect of having doula care covered/reimbursed by Medicaid for the relative month’s length (i.e. 3 months for those in “3”) of the pregnancy. A vertical dashed line is placed just before the first month of the policy taking effect, and another is placed at month 9. If effects are truly coming from the policy, the first month in which an effect could be detected would be month zero; if effects are coming from the policy but require the doula be available for most or all of a pregnancy, they should only be detected starting around month 9. Average post-period effects are shown in Table ??.

⁸Results including all births in the country can be generated, but require an intense amount of computing power and a lengthy time to process the data. Additionally, it is not clear that the non-Medicaid births, even within the same state, would provide a good counterfactual for Medicaid births, given the intensity of health disparities along income lines.

4 Results

In each set of results, the timing of doula care reimbursement is used as the source of exogenous variation in doula access. For the results to be valid, it must be the case that Medicaid-covered births in states that do not cover doula services must provide a good counterfactual for Medicaid-covered births in doula-covering states. The dynamic nature of the results (in all figures) captures whether the treatment effects differ by how much of a pregnancy occurred with Medicaid reimbursement for doula care as an option. Point estimates in the “3” slot represent the difference in outcomes for Medicaid births occurring 3 months after the state of residence moved to cover doula care, relative to other births; all compared to the omitted category of Medicaid-covered births that occurred one month prior to doula coverage. If simply having the option of the presence of a doula in labor is the driver of effects, we may anticipate all effects to be stagnant and consistent across all treated pregnancies; if having access to doulas prior to delivery matters, we might expect effects to increase with months of pregnancy occurring during coverage.

All non-Medicaid births in all states are dropped from these analyses, generating a comparison across states and time among Medicaid-covered birth events. Additionally, any state engaging in a doula initiative through a channel other than Medicaid is dropped from the analyses. This is a logical place to start, since Medicaid-covered births are more likely to be similar to other Medicaid-covered births than they are to the births of same-state births to privately-insured (and often wealthier) families. In this case, 0 represents births occurring the same month in which the birthing person’s state of residence began reimbursing doula care; 12+ represents births occurring 12 or more months after the state of residence began reimbursing doula care.

Visual results using the dynamic specification can be found in Figures 2 – ??, and the corresponding table of average difference-in-differences effects can be found in Table 4. The presented results are robust to the inclusion of demographic variables (mother’s age, marital status, paternity acknowledgement, child’s sex) and results for only first-time mothers are

similar for the full population. Generally speaking, results suggest improved health outcomes after Medicaid coverage of doula services, despite the fact that the doula takeup is low.

5 Conclusion

The current iteration of this paper shows that Medicaid coverage of doula care increases the presence of doulas among Medicaid-covered births shortly after the policy takes effect, which fade away with time. At its peak, the increase in doula presence for all birthing people may be as high as 30%, though estimates are noisy and fade quickly. Effects seem strongest for Asian American/Pacific Islander birthing people, with an average post-period increase of approximately 12%, but given the generally small use of doulas in delivery, this size of an effect is unlikely to have population-level impacts on observable infant and maternal health outcomes. That said, Medicaid births that have a doula present are generally healthier — with less Cesarean sections, less likelihood of early-term, low- or very low-birthweights, and higher likelihood of zero reported maternal morbidity indicators. Given the relatively low costs of supporting doula services, this work suggests that funding for certified doulas *could* improve outcomes while also saving states money.

6 Discussion

While this study certainly has room for expansions, it is clear that in providing coverage for doula services, some birthing people take up those services, and those that do seem to experience better birth-related health outcomes. Future work within this subject must better understand how the level or type of coverage/reimbursement structure matters, whether some groups or outcomes are more or less impacted by doula coverage, how supply of doulas changes in response to Medicaid reimbursement, and whether — for those truly most likely to be impacted by doula services — birth outcomes are changing in meaningful ways.

This project is being conducted alongside other work looking within states that begin to

cover doulas — using the variation in supply of doulas across counties. Additionally, this project will add analyses comparing within-state racial gaps in health outcomes as well as gaps in Medicaid-covered births relative to privately insured births.

Given the absolute tragedy that families and society experience with extreme poor birth-related outcomes (such as maternal or infant mortality), it is of the utmost importance for policy-makers to understand how to help create institutions that work toward reducing these terrible outcomes rather than increasing them. As labor and delivery units continue to close and reproductive health care continues to face legal battles — leading to physicians being unwilling or unable to provide at least some of the care that could prevent these deaths — the importance of addressing this crisis will only continue to rise.

Providing culturally-sensitive, trauma-informed care could be the difference between life and death for someone facing the physical, emotional, and financial journey of a pregnancy and delivery. Doulas could be an answer to this problem — and the costs of training and funding doulas is relatively small compared to other potential interventions.

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Tables

Table 1
Estimated Effects of Doula Coverage on Doula NPI Registrations

	New Registrations	Total Registrations
ATT	64.301*** (21.675)	240.488*** (62.460)
Observations	1386	1386

Notes: Results come from a model with state and year fixed effects.
Standard errors are in parentheses below the point estimates.

Table 2
Estimated Effects of Doula Coverage on Cesarean Sections

	All Births	AAPI	AIAN	Black	Hispanic	Multiracial	White
did=1	-0.0425*** (0.0022)	0.0312*** (0.0097)	-0.0580** (0.0295)	-0.0497*** (0.0049)	-0.0801*** (0.0042)	-0.0208 (0.0157)	-0.0356*** (0.0026)
Observations	41254523	2322492	389130	6292558	10031485	1032228	29929799

Notes: Results come from a model with county and year fixed effects and uses population weighting.

Standard errors are in parentheses below the point estimates.

Table 3
Estimated Effects of Doula Coverage on Vaginal Deliveries

	All Births	AAPI	AIAN	Black	Hispanic	Multiracial	White
did=1	0.0425*** (0.0022)	-0.0312*** (0.0097)	0.0580** (0.0295)	0.0497*** (0.0049)	0.0801*** (0.0042)	0.0208 (0.0157)	0.0356*** (0.0026)
Observations	41254523	2322492	389130	6292558	10031485	1032228	29929799

Notes: Results come from a model with county and year fixed effects and uses population weighting.

Standard errors are in parentheses below the point estimates.

Table 4
Estimated Effects of Doula Coverage on Failed Labor Attempts

a1	a2	a3	a4	a5	b/se	b/se	b/se	b/se	b/se	faillabor	1.did	-0.0607***	0.0077	0.0182	-0.0855***	-0.1629***	(0.0037)	(0.0150)	(0.0486)	(0.0082)	(0.0075)	O
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Notes: Results come from a model with county and year fixed effects and uses population weighting.

Standard errors are in parentheses below the point estimates.

Table 5
Estimated Effects of Doula Coverage on Preterm Births

	All Births	AAPI	AIAN	Black	Hispanic	Multiracial	White
did=1	-0.0552*** (0.0031)	-0.0635*** (0.0149)	-0.0127 (0.0397)	-0.0371*** (0.0064)	-0.1194*** (0.0061)	-0.0567** (0.0225)	-0.0569*** (0.0039)
Observations	41255286	2322501	388599	6290675	10031319	1032008	29927420

Notes: Results come from a model with county and year fixed effects and uses population weighting.

Standard errors are in parentheses below the point estimates.

Table 6
Estimated Effects of Doula Coverage on Low Birth Weight

	All Births	AAPI	AIAN	Black	Hispanic	Multiracial	White
did=1	-0.0219*** (0.0037)	-0.0201 (0.0162)	-0.0878* (0.0502)	-0.0109 (0.0070)	-0.0389*** (0.0077)	-0.0027 (0.0260)	-0.0218*** (0.0048)
Observations	41278712	2323405	389236	6295426	10037407	1032714	29943301

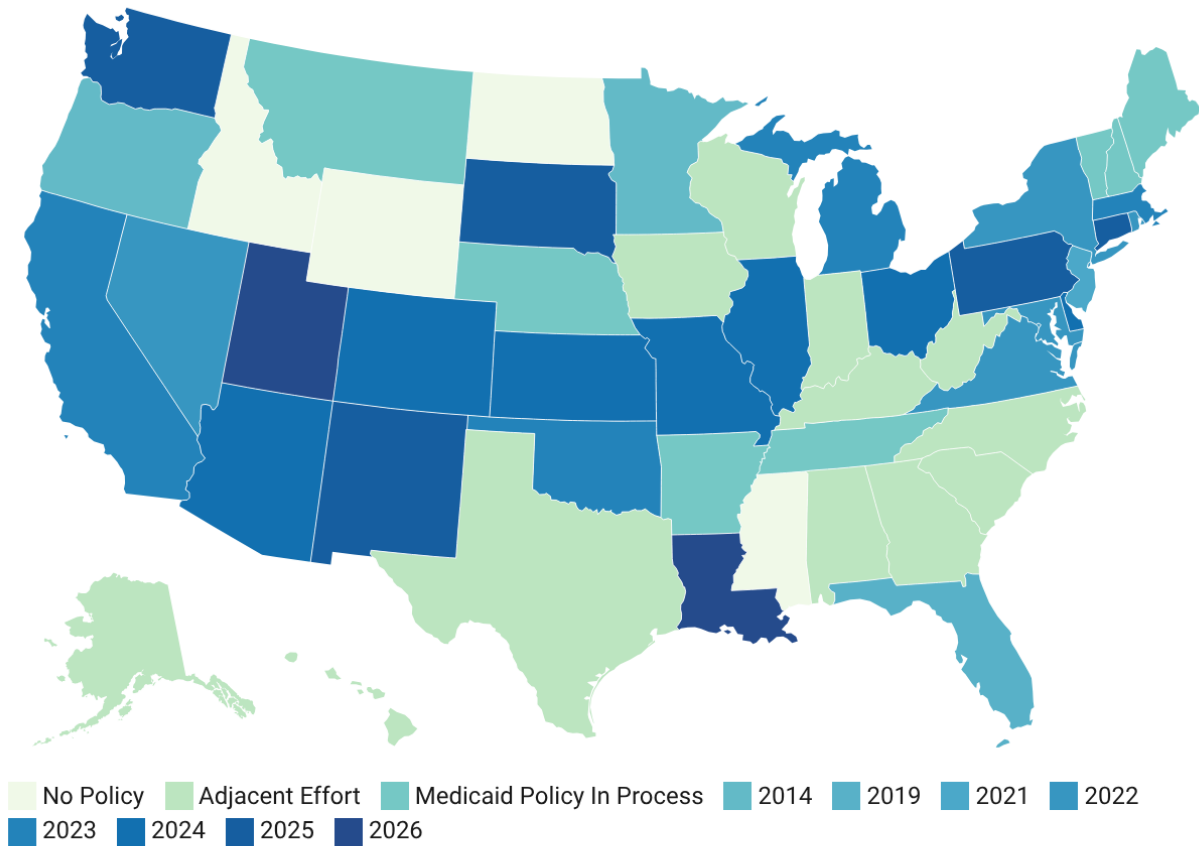
Notes: Results come from a model with county and year fixed effects and uses population weighting.

Standard errors are in parentheses below the point estimates.

Figures

Figure 1
States with Doula Programs

Status of State Medicaid Reimbursement for Doulas

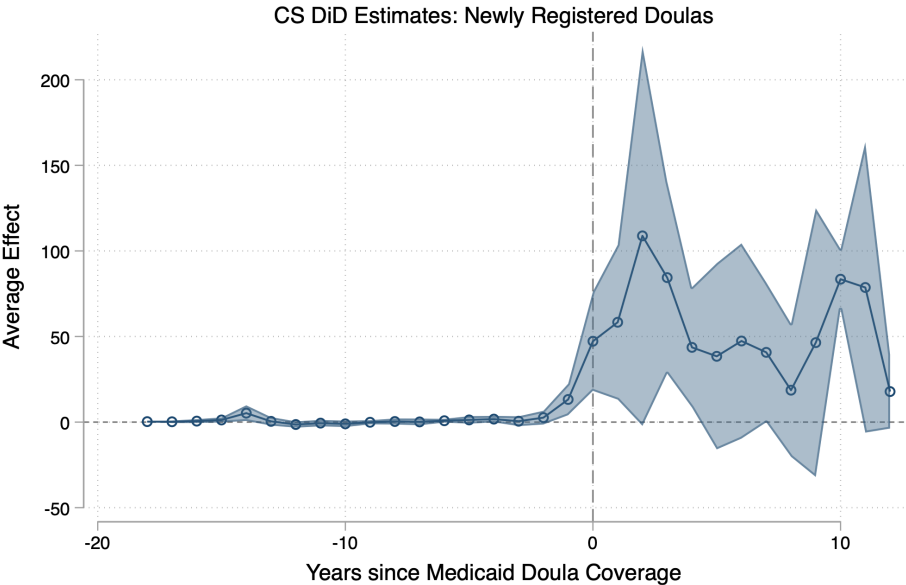


Created with Datawrapper

Notes: This map, built with DataWrapper, presents the timing of the start of state Medicaid doula reimbursements.

Figure 2
Effects on Doula NPI Registries

New Doula Registries



Total Doula Registries

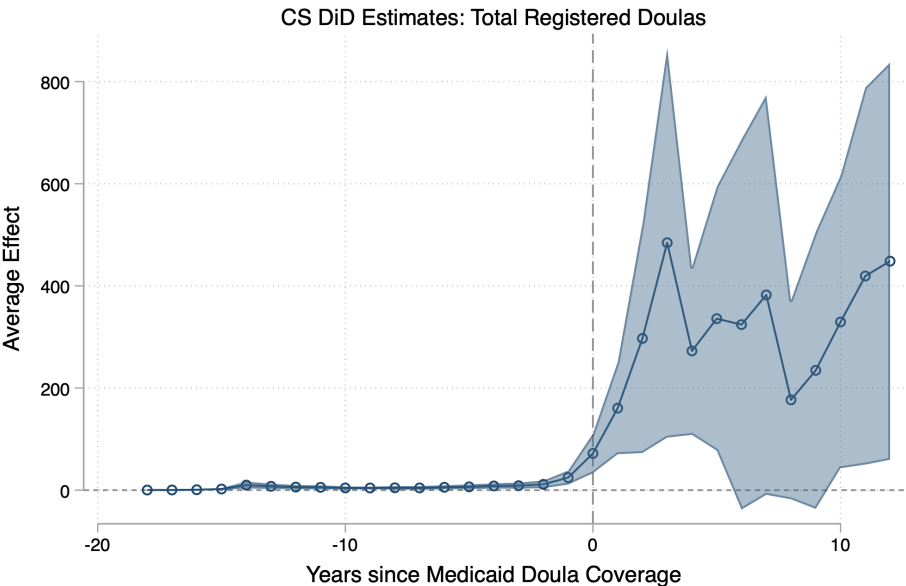


Figure 3
Effects on Failed Labor Attempts – LPM



Notes:

Figure 4
Effects on Failed Labor Attempts – Logit

